

IPTV is also a multicast and unicast platform, through which viewers can choose from hundreds of channels. They can also demand for channels of their choice in IPTV, whereas cable TV and satellite TV are only broadcast modes, wherein viewers can see only what is being broadcast. This distinct differentiator makes IPTV unique and desirable. IPTV requires a broadband connection, an STB for access and subscriber identity to secure content. It can be viewed on a normal TV or PC.

However, with IPTV, one question comes up frequently: Where does the STB receive its pictures from? "Most video content enters the system at the telco's national head-end, where network feeds are pulled from satellites and encoded if necessary. The video stream is broken up into IP packets and put into the telco's core network, which is a massive IP network that handles all sorts of traffic (data, voice, etc), in addition to the video," explains Sunder Raju, managing director, ACT Television.

The video streams are received by a local office, where local content (such as TV stations, advertising and VoD) is appended to the mix. It's also the spot where the IPTV middleware is housed. This software stack handles user authentication, channel change requests, billing and VoD requests.

How does an IPTV subscriber get hundreds of channels with a digital subscriber line (DSL)? "When a user changes the channel on his STB, the box does not tune a channel like a cable system. It switches channels by using the IP group membership protocol (IGMP) v2 to join a new multicast group. When the local office receives this request, it checks to make sure that the user is authorised to view the new channel, then directs the routers in the local office to add that particular user to the chan-

IPTV Vs Digital and Cable TV

Feature	Cable	DTH	IPTV
Live channel	✓	✓	✓
Time-shift TV	×	×	✓
Video-on-demand	×	Limited	Instant
Interactive movies	×	×	✓
Rewind/FF/pause movies	×	×	✓
Replay live channels	×	×	✓
Interactive CAS	×	×	✓
Interactive gaming	×	×	✓
Messaging	×	×	✓
Video classifieds	×	×	✓
Reception quality	Average	Good	Best
Triple play	×	×	✓
Works in heavy rain	×	×	✓

nel's distribution index. In this way, only signals that are currently being watched are actually being sent from the local office to the digital subscriber line access multiplexer (DSLAM) and finally, to the user," informs Choudhary.

IPTV vs DTH, digital/cable TV

A common question is being often asked: Does IPTV score over digital TV, cable TV and DTH? "It will not be appropriate to compare IPTV with other modes of broadcast. IPTV as a service will further augment what is already available on DTH and other traditional

forms. Thus, IPTV will complement them and not compete with them," says Himanshu Sahu, assistant director, sales engineering, IPTV & VAS, UTStarcom, India. "However, there are several unique advantages of IPTV that keep it miles ahead of other traditional TV delivery mechanisms," he adds.

According to Prakash, "The entry cost of IPTV service is at par with cable TV but DTH and digital TV require much more bandwidth (8 Mbps per home, per connection), whereas IPTV can be viewed at 2-3 Mbps."

Advantages galore

"IPTV is a truly interactive service and can provide numerous services like e-commerce, e-governance, e-education, e-medicine, weather and information, etc. However, its penetration is heavily dependent on good quality, high bandwidth and broadband connectivity. The video tier has to be managed by a content delivery network (CDN), which sits on top of the telcos' existing telecom network," points out Prakash.

IPTV is still in its nascent phase in India. Considering the current scenario, IPTV can be said to have both pros and cons.

Interactive nature. One of the best advantages of IPTV is that it uses Internet protocol to provide two-way communication, making the television viewing experience truly interactive. It opens the door



ACT Interactive's IPTV STB & remote